

1 Epidemic Modelling Intro

2 Epidemic Modelling; Intro to LaTeX and R



Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 4MB3/6MB3 Mathematical Biology

Instructor: David Earn

Lecture 1
Epidemic Modelling Intro
Tuesday 8 September 2026

Course information

- Course web page: <http://ms.mcmaster.ca/earn/4MB3>
 - Click on “Course information”.
 - Let's have a look now...

Mathematical Biology Research Seminar (MBRS)

Most weeks, there is a Mathematical Biology Research Seminar, which you are encouraged to attend if you are available.

- Where: **TBA**
- When: **Thursdays, 2:30 p.m.**
- Starts: **TBA**

If you would like to be on the e-mail distribution list for these seminars, please send me an e-mail (with MBRS in the subject line).

In-class polls: a form of participation

I find these work well, but with such a small class I think we'll just stick to in-class discussion.

Group formation

Most work in this course will be done in groups.

- Form a group of 2 or 3 students **during the break TODAY.**
- **Exactly one** member of your group must e-mail the instructor **TODAY during the break:**
 - Include “Math 4MB3” and your proposed group name in the subject line.
 - **Copy your message to all members of your proposed group so I have everyone’s e-mail in the thread.**
- After I receive the group-name e-mails, I will send each student an e-mail invitation to the Background and Group Formation Survey.
- Note: *Instructor may change groups based on survey results.*

Online Surveys

*You will be required to fill in online surveys during this course.
Doing so in a timely manner contributes to your participation mark.*

The first online survey is the Background and Group Formation Survey:

- I will e-mail invitations after I have received the group-name e-mails.
- Please complete the survey as soon as possible after receiving the invitation.
- It should take only ~ 5 minutes.
- Note that *surveys sometimes fail to save*.
 - Type long answers into a file first and paste them into the survey. Then you won't get as frustrated if it fails to save.

Software

- **ASAP**, install the software discussed on the [Software page](#) on the [course web site](#):

- L^AT_EX



- R



- RStudio



- R packages of possible interest:
epigrowthfit, fastbeta, fitode.

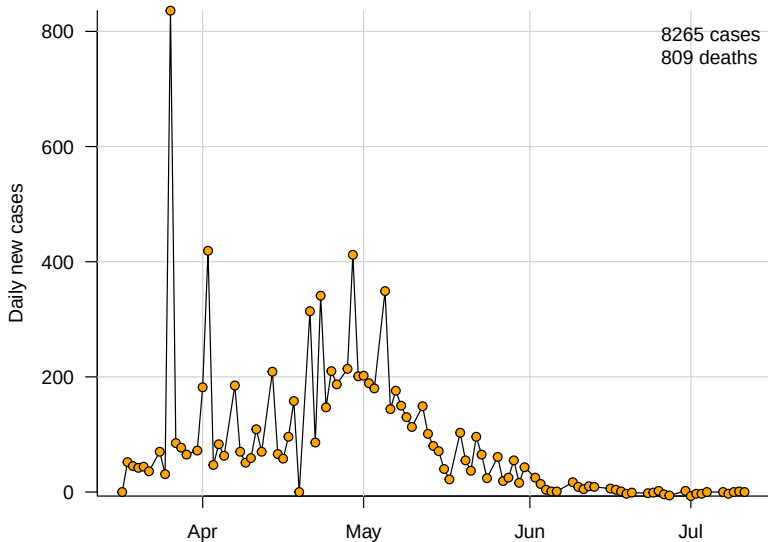
- XPPAUT

- Emacs

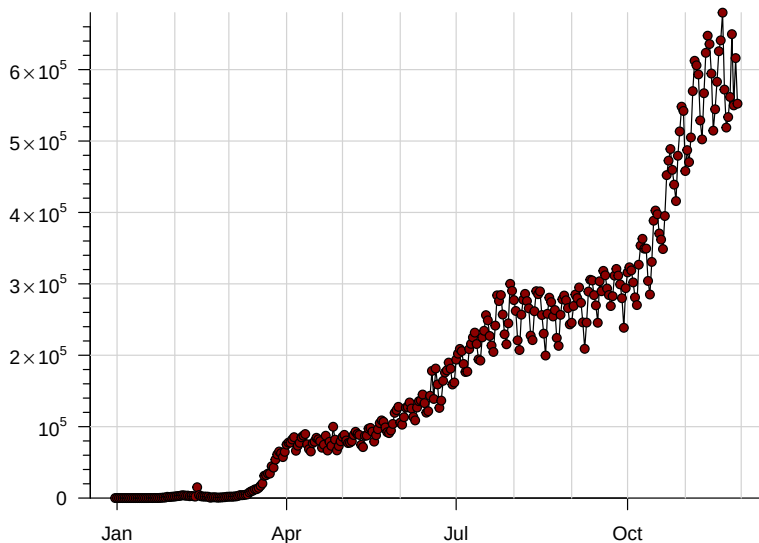
- *Note*: the [Software page](#) also contains some info about spell-checking and counting words in L^AT_EX documents.

Epidemic Modelling

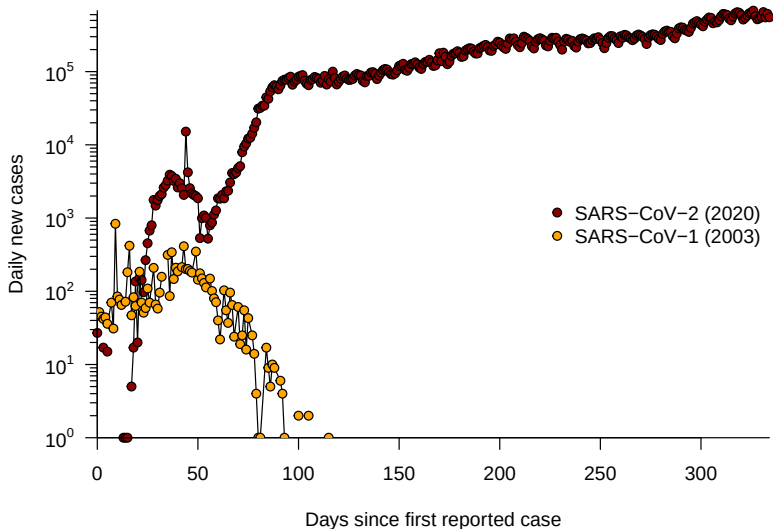
Daily SARS-CoV-1 in 2003 (Worldwide)



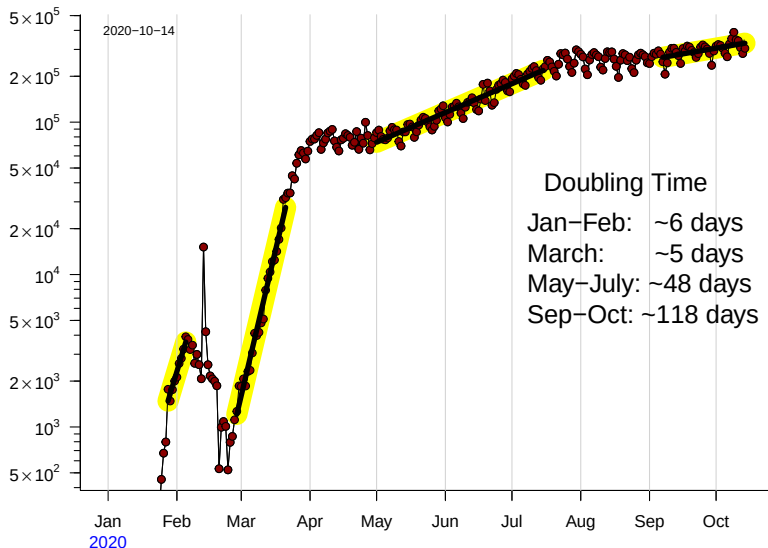
Daily SARS-CoV-2 in 2020 (Worldwide)



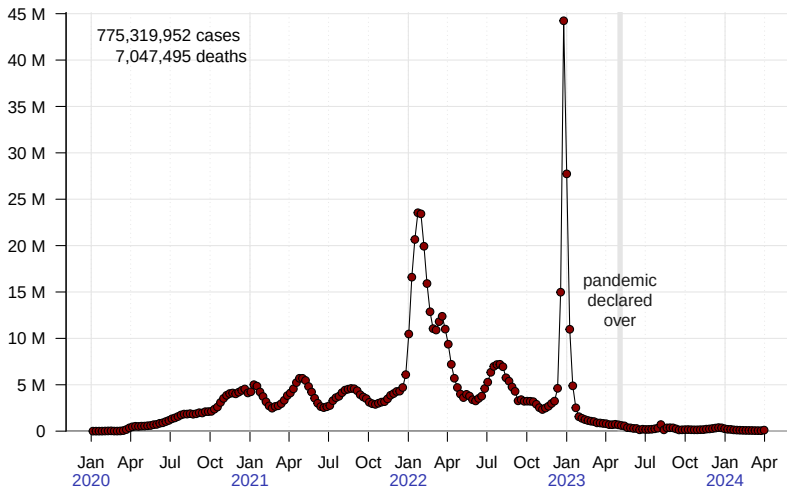
Daily SARS-CoV-1 vs SARS-CoV-2 (Worldwide)



Daily SARS-CoV-2 (Worldwide) exponential growth fits

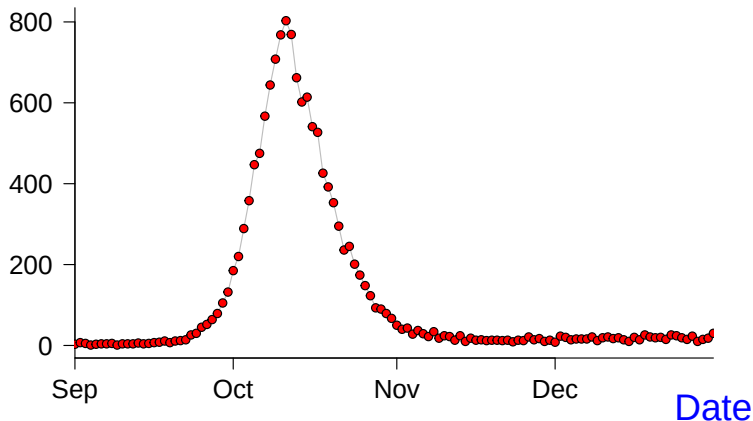


Weekly SARS-CoV-2 (Worldwide) for the first four years



Pneumonia & Influenza Mortality, Philadelphia, 1918

P&I Deaths



Modelling challenge

Develop a model that helps us understand the [graph on the previous slide](#), based on mechanisms of disease spread.

- Only one variable is observed (P&I deaths per day) so construct a model containing only one variable.
- Think about how disease spreads and express your thoughts with mathematical notation.
- Derive a differential equation that models the process of disease transmission.
- Analyze the model and determine its strengths and weaknesses/limitations.

Make (Biological) Assumptions Clear

- 1 *Assume* the disease is transmitted by contact between an infected individual and a susceptible individual.
- 2 *Assume* the latent period (delay between being infected and becoming infectious) is so short that it can be ignored (technically assume it is zero).
- 3 *Assume* all members of the population are identical and respond identically to the disease. In particular, all susceptible individuals are equally susceptible and all infected individuals are equally infectious.
- 4 *Assume* the population size is fixed during the epidemic, *i.e.*, ignore births, migration, and deaths from causes other than the disease, and count individuals who have died from the disease as part of the population.

About Assumptions...

- Note that the first assumption on the [previous slide](#) is actually correct.
- The other assumptions are wrong, but are reasonable approximations.
- It is best to *start as simple as possible and add complexity later*, in order to:
 - obtain a model that actually succeeds in explaining [the data](#) with as few ingredients as possible;
 - check that inferences we draw from our model(s) are robust to the inclusion of more biological details/realism.

What variables should we include in our model?

- Independent variable: time (t)
- Dependent variable: Many options, e.g.,
 - Incidence (number of new infections per unit time)
 - Prevalence (total current number of infected individuals)
 - Death rate (number of deaths per unit time)
 - Death toll or cumulative deaths (number of deaths so far)
- So, what would be best?
- Not deaths, because whether or not you die may be unrelated to how much you transmit.
- But deaths are what **we observe**! What to do?!?
- Make another assumption...

Additional assumption(s)

- We actually want to know incidence or prevalence, but we **observe deaths**.
- Under what circumstances would daily deaths be a good estimate of incidence? (*i.e.*, What must we assume in addition to the **assumptions we have already made**.)
 - 5 **Assume** that the time from infection to death is exactly the same (a certain number of days) for every individual who dies.
 - 6 **Assume** that the probability of dying from the disease is the same for every individual who is infected.
- Then daily death counts are proportional to daily incidence a certain number of days in the past, *i.e.*, the “mortality curve” that **we observe** is a translated and scaled version of the “epidemic curve” (new cases per day).

So...what variables should we include in our model?

- Independent variable: time (t)
- Dependent variable: one of:
 - Incidence (number of new infections per unit time)
 - Prevalence (total current number of infected individuals)
- Which one?
- Choose prevalence (I) because anybody who is currently infectious can infect others, so it will probably be easier to formulate a transmission model based on prevalence.
(Try not to lose sight of underlying biological mechanisms.)
- But our **mortality curve** is related to incidence, not prevalence!?! Argh. What to do?!?
- Let's work with prevalence and see how it works out.
Maybe we'll be able to derive the incidence curve from a model based on prevalence.

Notational note

- We use $/$ for prevalence because prevalence is the number of infected individuals.
- So, let's try to write down a model...

A first (naïve) attempt at an epidemic model

- Variables: time t , prevalence $I(t)$
- How does I increase?
- Start with I_0 infected individuals at time $t = 0$. What happens for $t > 0$.
- Let B = average number of contacts with susceptible individuals that lead to a new infective per unit time per infective in the population (and suppose B is constant). Then

$$I(t + \Delta t) \simeq I(t) + B I(t) \Delta t$$

- In the limit $\Delta t \rightarrow 0$, we have

$$\frac{dI}{dt} = B I \quad \implies \quad I(t) = I_0 e^{Bt}$$

Beware: implicit assumptions that should be explicit

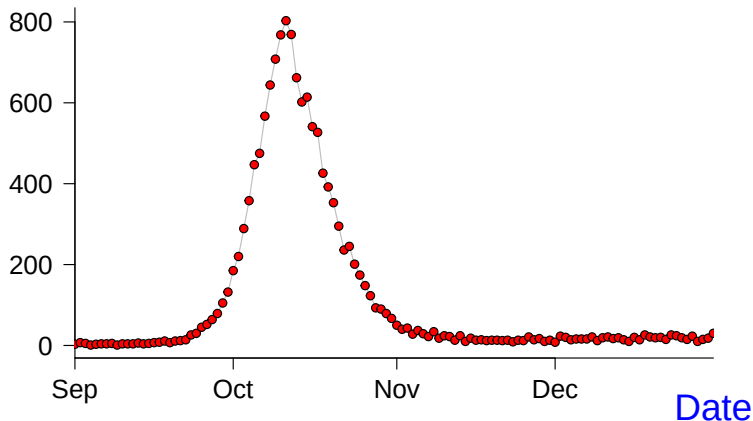
- Ignored discrete nature of individuals when taking limit.
- *Ignored finite infectious periods!*
 - Sometimes it isn't obvious that we've made some assumptions until after we see what the model predicts.

How can we tell if our model is good?

- Compare model predictions with **data**.
- What is the best way to do that?
- Depends on what predictions we're trying to test.
- Model predicts exponential growth.
How should we test that prediction?
- Transforming **the data** might help.

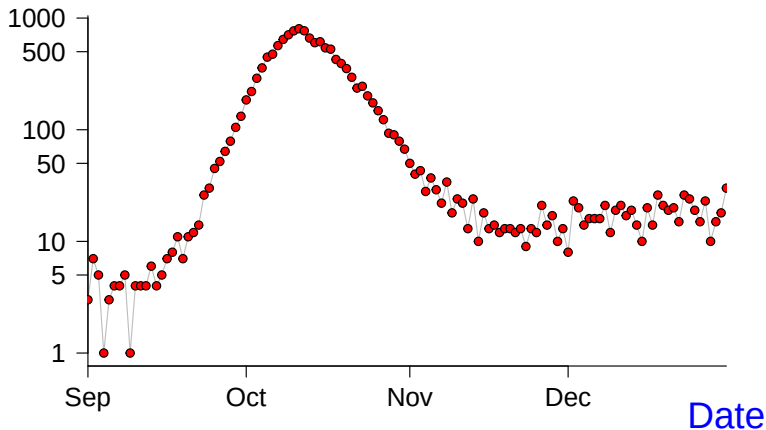
Original data: P&I Mortality, Philadelphia, 1918

P&I Deaths



Logarithmic scale: P&I Mortality, Philadelphia, 1918

P&I Deaths



Parameter estimation

How can we estimate the model parameters, I_0 and B , from the $P&I$ data?

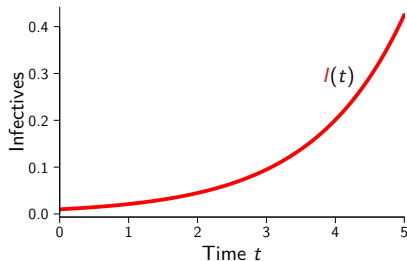
- Fit a straight line through the part of the **logarithmic mortality curve** that looks straight.
- The slope of the line is B .
- The “intercept” is $\log I_0$.
 - “Intercept” in quotes because we need to define $t = 0$ as the time when exponential growth begins.
- **Note:** Parameter estimation is, in general, a very tricky business and deserves a great deal of attention.

Earn, Park, Bolker (2024) “Fitting Epidemic Models to Data...” *Bull. Math. Biol.* **86**, 109

Naïve epidemic model

- Variables: time t , prevalence $I(t)$
- Parameter B = average number of contacts with susceptible individuals that lead to a new infective per unit time per infective in the population

$$\frac{dI}{dt} = BI \implies I(t) = I_0 e^{Bt}$$



Naïve model: the good and the bad

■ Good:

- Makes clear predictions
- Predictions can be tested
- Estimation of parameter (B) is easy
 - B is the slope of the straight portion of the epidemic curve on the log scale. (Why?)
 - Remember we are imagining that the **mortality curve** is equivalent to the epidemic curve after translation and scaling.
 - Why do translation and scaling not affect the estimate of B ? Assignment 2...

■ Bad:

- Model is consistent only with exponential growth phase.
- Absurd long-term prediction: unbounded growth in $I(t)$
 - Implicitly assumed that population size $N = \infty$.

How can we improve our model?

- Insist that population size is finite ($N < \infty$).
- Keep track of both **infectives** $I(t)$ and **susceptibles** $S(t)$.
- Assume individuals who are *not infected* are **susceptible**:

$$I(t) + S(t) = N = \text{constant.}$$

New model parameter(s)?

- B = average number of contacts with **susceptible** individuals that lead to a new **infective** per unit time per **infective**
- In the naïve model, we assumed $B = \text{constant}$.
Is B really constant?
- B depends on how many **susceptibles** there are.
- $B = \beta S(t)$
- β = average number of contacts between **susceptibles** and **infectives** that lead to a new **infective**
per unit time
per **infective**
per **susceptible**
- β is called the ***transmission rate***.

Revised epidemic model: “SI model”

$$\frac{dI}{dt} = \beta S(t)I(t)$$

- Two state variables. One equation. Problem? No:

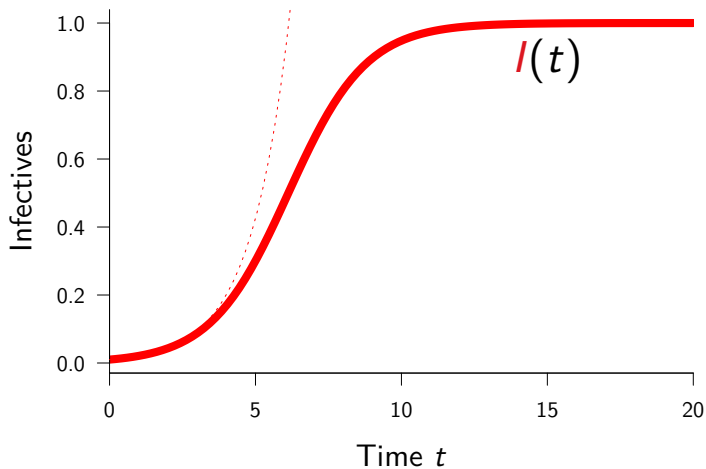
$$\frac{dS}{dt} = -\beta S(t)I(t)$$

- But $S(t) = N - I(t) \implies I(t)$ is still the only variable:

$$\frac{dI}{dt} = \beta I(N - I)$$

- Is this a better model?
- What does it predict?

SI model: Example solution



SI model: Analysis

$$\frac{dI}{dt} = \beta I(N - I)$$

- We can find the exact solution. How? (Assignment 1)

$$I(t) = \frac{I_0 e^{N\beta t}}{1 + (I_0/N)(e^{N\beta t} - 1)}$$

- But exact solution is not particularly enlightening.
- Qualitative analysis:
 - Initially $I \ll N$. What does the model predict then?
Exponential growth. Great!
 - As I grows, growth rate slows. Why?
Fewer and fewer **susceptibles** to infect.
 - Asymptotic behaviour? Equilibria? Periodic orbits?
(periodic orbit = recurrent epidemics)
 - (Non-trivial) periodic orbits impossible in one dimension
(existence-uniqueness theorem).
 - Consider equilibria...

SI model: Equilibrium Analysis

$$\frac{dI}{dt} = \beta I(N - I), \quad I \in [0, N]$$

- Two equilibria:
 - $I = 0$ Disease Free Equilibrium (**DFE**)
 - $I = N$ Endemic Equilibrium (**EE**)
- Stability:
 - DFE is unstable ($0 < I < N \implies dI/dt > 0$)
 - EE is locally asymptotically stable (**LAS**)
 - EE is globally asymptotically stable (**GAS**)
(stability of EE follows from $0 < I < N \implies dI/dt > 0$)
(GAS requires a little more analysis... Assignment 1)
 - *Note:* In one dimension, global analysis always easy.
In higher dimensions, often try to find Lyapunov function.
(Lyapunov function for EE of SI model?... Assignment 1)
- Conclusions identical for any $\beta > 0$.

SI model: Biological Inferences

- For *any* transmission rate β :
 - Initially, exponential growth of cases.
 - Eventually, convergence to equilibrium (EE) at which *everyone* in the population is **infective**. hmmm...
- Is this model better than our first naïve model?
YES.
 - Still correctly predict initial exponential growth.
 - Can match epidemic curve for longer (up to the peak).
 - Does not predict absurd unbounded growth in **infective** population.
 - But this model cannot explain the decline of the epidemic.
- What should we do? Two obvious options:
 - 1 Get depressed, drop the course.
 - 2 Try to improve the model.

Recall motivating data: 1918 flu in Philadelphia

- Mortality curve (linear scale)
- Mortality curve (log scale)

How can we improve on the SI model?

- Include a key biological fact:
Individuals do not stay infectious with flu forever
- Either they *recover* and are immune thereafter, or they *die* (doesn't matter which) (well, maybe to them it does)
- Why don't we care if someone recovers or dies?
(i.e., Why doesn't it affect our dynamical inferences?)
- Because in either case the individual is *removed* from the transmission process, hence cannot affect the future pattern of the epidemic.

The SIR model

Introduce new class of **removed** individuals:

- $R(t)$ = number of individuals who have either recovered and are now immune or have died
- Let γ = rate of removal from the **infective** class (via recovery or death)

$$\frac{dS}{dt} = -\beta SI$$

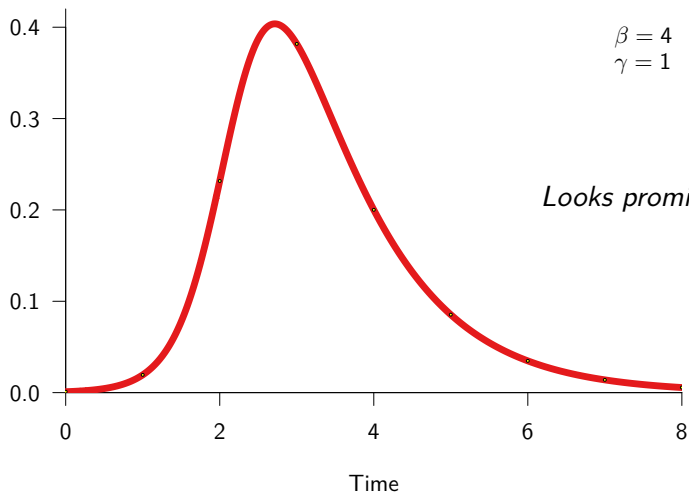
$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = \gamma I$$

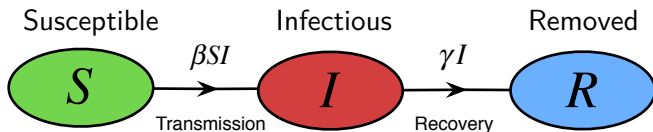
- Note: $\frac{dS}{dt} + \frac{dI}{dt} + \frac{dR}{dt} = 0 \implies S + I + R = N = \text{constant}$
- Convenient to rescale variables by N and interpret S, I, R as *proportions* of the population in each disease state.

The SIR model: Example numerical solution

Proportion infected $I(t)$



The SIR model: Flow Chart and Parameters



$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = \gamma I$$

■ Parameters:

- Transmission rate β
- Recovery rate γ
(or Removal rate)

The SIR model: Derived Parameters



$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = \gamma I$$

■ Derived Parameters:

- Initial growth rate $\beta - \gamma$
- Mean infectious period $\frac{1}{\gamma}$
- Basic Reproduction Number

$$\mathcal{R}_0 = \frac{\beta}{\gamma}$$

The SIR model: Derived parameters

The initial growth rate

- How do we calculate the initial growth rate from our model?
- Consider change in prevalence initially (when $I \ll 1$):

$$\begin{aligned}\frac{dI}{dt} &= \beta SI - \gamma I \\ &= (\beta S - \gamma) I \\ &\approx (\beta - \gamma) I \quad \text{if } S \sim 1 \text{ initially.}\end{aligned}$$

- $\therefore$ Initially $I(t) \approx I_0 e^{(\beta - \gamma)t}$.
- $\therefore$ Initial slope of logged prevalence curve is $\beta - \gamma$.

The SIR model: Derived parameters

The mean infectious period

- How do we calculate the mean infectious period from our model?
- Suppose at time 0 there are I_0 infectious individuals, and suppose we can *prevent contact with susceptibles*.
- The equation for I then simplifies to

$$\frac{dI}{dt} = -\gamma I, \quad I(0) = I_0$$

- We can solve this immediately to find

$$I(t) = I_0 e^{-\gamma t}$$

The SIR model: Derived parameters

The mean infectious period, continued...

- Thus, after time t , the number of people still infectious is reduced by a factor $e^{-\gamma t}$
- *i.e.*, the proportion of individuals who have an infectious period shorter than t is $1 - e^{-\gamma t}$
- *i.e.*, the cumulative distribution of the infectious period is $C(t) = 1 - e^{-\gamma t}$.
- Therefore, the probability density of the infectious period is $p(t) = C'(t) = \frac{d}{dt}(1 - e^{-\gamma t}) = \gamma e^{-\gamma t}$
- The mean of this exponential probability distribution is $\int_0^\infty t p(t) dt = \int_0^\infty t \gamma e^{-\gamma t} dt = \frac{1}{\gamma}$

The SIR model: Derived parameters

The basic reproduction number $\mathcal{R}_0$

$$\begin{aligned}\mathcal{R}_0 &= \beta \cdot \frac{1}{\gamma} \\ &= (\text{transmission rate}) \\ &\quad \times (\text{mean infectious period})\end{aligned}$$

- $\mathcal{R}_0$ is dimensionless
- $\mathcal{R}_0$ is the average number of secondary cases caused by a primary case (in a wholly susceptible population).
- We must have $\mathcal{R}_0 > 1$ to have an epidemic. Why?
 - $\frac{dI}{dt} = \beta SI - \gamma I = (\mathcal{R}_0 S - 1)\gamma I$
 - $\therefore \mathcal{R}_0 \leq 1 \implies \frac{dI}{dt} \leq 0$ for all $(S, I) \in [0, 1]^2 \implies$ no growth

The SIR model: Analysis (biological well-posedness)

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

- We need S , I and R all non-negative at all times.

- Does $0 \leq S(0) + I(0) \leq 1$ imply $0 \leq S(t) + I(t) \leq 1$ for all $t > 0$?

- $S = 0 \implies S' = 0$, so
 $S(0) \geq 0 \implies S(t) \geq 0 \forall t > 0$.

- $I = 0 \implies I' = 0$, so
 $I(0) \geq 0 \implies I(t) \geq 0 \forall t > 0$.

- $(S + I)' = S' + I' = -\gamma I \leq 0$
 $\implies S + I$ is always non-increasing
 $\implies S(t) + I(t) \leq S(0) + I(0) \leq 1$.

**Be careful:
Is this a sensible
biological
model?**

The SIR model: Analysis (equilibria etc.)

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

■ Equilibria:

$(S, I) = (S_0, 0)$ for any $S_0 \in [0, 1]$

- Continuum of equilibria.
- Does this make sense?
- Biological meaning of equilibria?

■ Linearization:

$$DF_{(S,I)} = \begin{pmatrix} -\beta I & -\beta S \\ \beta I & \beta S - \gamma \end{pmatrix}$$

$$DF_{(S_0,0)} = \begin{pmatrix} 0 & -\beta S_0 \\ 0 & \beta S_0 - \gamma \end{pmatrix}$$

- All equilibria are *non-hyperbolic*.

■ Periodic orbits:

- Are there periodic solutions?

$$(S + I)' = -\gamma I$$

$\implies$ no periodic orbits. Why?

- If $I(0) = 0$ then equilibrium.
- If $I(0) > 0$ then $(S + I)' < 0$, so cannot increase back to initial state.
- Also follows from [Index Theorem](#) (cannot enclose any equilibria).

The SIR model: Analysis

Nullclines:

- $S' = 0 \implies S = 0$ or $I = 0$
 - S nullclines: both coordinate axes

$$\frac{dS}{dt} = -\beta SI$$

- $I' = 0 \implies I = 0$ or $S = \gamma/\beta$
 - I nullclines: S axis and vertical line at $S = 1/\mathcal{R}_0$
 - Is the I nullcline at $S = 1/\mathcal{R}_0$ always relevant?
 - If, and only if, $\mathcal{R}_0 > 1$.
 - If $\mathcal{R}_0 < 1$ then $S = 1/\mathcal{R}_0$ is outside the biologically relevant region of the (S, I) phase plane.

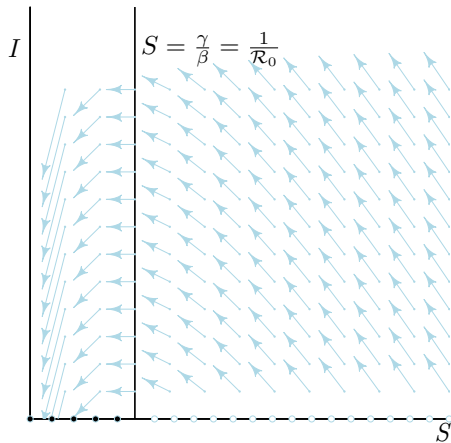
$$\frac{dI}{dt} = \beta SI - \gamma I$$

The SIR model: Analysis

Nullclines and Direction Field ($\mathcal{R}_0 = 4$):

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$



The SIR model: Analysis

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

Phase Portrait:

- We cannot find solutions $S(t)$ and $I(t)$ for this system.
- We *can* find exact analytical solution for the phase portrait!



Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 4MB3/6MB3 Mathematical Biology

Instructor: David Earn

Lecture 2

Epidemic Modelling; Intro to LaTeX and R
Tuesday 15 September 2026

Announcements

- [Groups](#).
- [Assignment 1](#) is due on Thursday 24 September 2026 at 11:59 p.m. on [Crowdmark](#).
- Have you successfully installed latex? R? `epigrowthfit`? `fastbeta`? `XPPAUT`? Emacs?

The SIR model: Analysis

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

Phase Portrait:

- We cannot find solutions $S(t)$ and $I(t)$ for this system.
- We *can* find exact analytical solution for the phase portrait!

- i.e., we can find an expression $I(S)$ for solution curves in the (S, I) phase plane.
- Slope of $I(S)$ depends only on S :

$$\frac{dI}{dS} = \frac{dI/dt}{dS/dt} = -1 + \frac{1}{\mathcal{R}_0 S} \quad (*)$$

- Note: Slope is flat for $S = 1/\mathcal{R}_0$, so max or min of $I(S)$ occurs on I nullcline if $\mathcal{R}_0 > 1$
- Easy to integrate (*):

$$\int_{I_0}^I dI = \int_{S_0}^S \left(-1 + \frac{1}{\mathcal{R}_0 S}\right) dS$$
- $I - I_0 = -(S - S_0) + \frac{1}{\mathcal{R}_0} \log(S/S_0)$

The SIR model: Analysis

Model Equations:

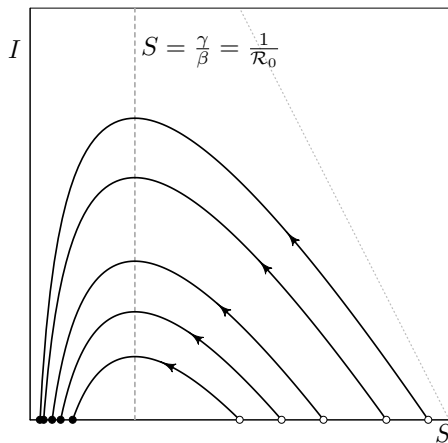
$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

Solution Curves in Phase Plane:

$$\begin{aligned} I + S - (I_0 + S_0) \\ = \frac{1}{\mathcal{R}_0} \log(S/S_0) \end{aligned}$$

Phase Portrait ($\mathcal{R}_0 = 4$):



The SIR model: Analysis

Model Equations:

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

Solution Curves in Phase Plane:

$$\begin{aligned} I + S - (I_0 + S_0) \\ = \frac{1}{\mathcal{R}_0} \log(S/S_0) \end{aligned}$$

Final Size of Epidemic:

- As $t \rightarrow \infty$ we have $(I_\infty + S_\infty) - (I_0 + S_0) = \frac{1}{\mathcal{R}_0} \log S_\infty/S_0$
- But for a newly invading pathogen: $S_0 \simeq 1, I_0 \simeq 0, I_\infty = 0$
- In the limit $I_0 \rightarrow 0$, we have $(S_\infty - 1) = \frac{1}{\mathcal{R}_0} \log S_\infty$
- Define “Final Size” $Z = 1 - S_\infty$
- $\therefore -Z = \frac{1}{\mathcal{R}_0} \log(1 - Z)$, i.e.,

$$Z = 1 - e^{-\mathcal{R}_0 Z}$$

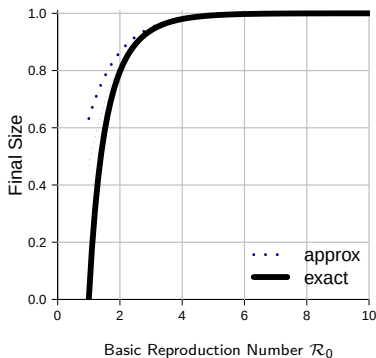
- This is a famous formula, derived by Kermack and McKendrick in 1927.

The SIR model: Analysis

Final Size Formula:

$$Z = 1 - e^{-\mathcal{R}_0 Z}$$

- Final size is determined entirely by $\mathcal{R}_0$
- Final size is never the whole population ($Z < 1$)
- Formula is valid for much more realistic models (Ma & Earn, 2006)



- For 1918 flu: $1.5 \lesssim \mathcal{R}_0 \lesssim 2$
- Proportion of world population infected? $\sim 60\text{--}80\%$
- For COVID-19: $3 \lesssim \mathcal{R}_0 \lesssim 15$ (depends on variant)

From Final Size to Reproduction Number

- The **final size relation** allows us to estimate the proportion of the population that will be infected *given* an estimate of $\mathcal{R}_0$.
- But we can turn it around: if we know the **final size** Z then we can easily estimate $\mathcal{R}_0$:

$$Z = 1 - e^{-\mathcal{R}_0 Z} \quad \Rightarrow \quad \mathcal{R}_0 = -\frac{1}{Z} \log(1 - Z)$$

- This is useful *post-hoc* only (*after* an epidemic) and only if $\mathcal{R}_0$ did not change during the epidemic.

The SIR model: Non-dimensionalization

- Often helpful to use dimensionless parameters.
- How do we identify “the right” dimensionless parameters?
- $\frac{\beta}{\gamma}$? $\frac{\gamma}{\beta}$? $\frac{\beta}{\beta+\gamma}$? $\frac{\beta^2}{\beta^2+\gamma^2}$?
- We choose β/γ because it has a natural interpretation.
- But we are still left with γ as a second parameter.
- Can we simplify the model somehow?
- γ defines a time scale ($1/\gamma$ is the mean infectious period).
- If time unit is mean infectious period, then $\gamma = 1$.
- So in these “natural” time units, the SIR model is

$$\frac{dS}{dt} = -\mathcal{R}_0 S I, \quad \frac{dI}{dt} = \mathcal{R}_0 S I - I$$

- There is really only one parameter in the model. The other is just a time scale and does not affect the *qualitative* dynamics.

The SIR model: Results so far

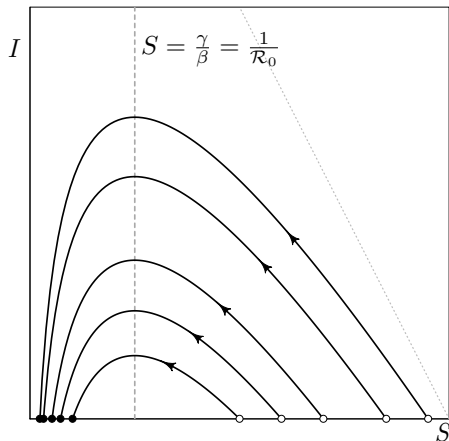
Mathematical Results:

- Model is biologically well-posed
 - $0 \leq S(0) + I(0) \leq 1 \implies 0 \leq S(t) + I(t) \leq 1 \quad \forall t > 0$
- No periodic orbits.
- Continuum of equilibria.
- Stability of equilibria:
 - Linearization useless (all equilibria non-hyperbolic).
 - Further analysis necessary.
- Exact solution for phase portrait:

$$I(S) = I_0 + (S_0 - S) + \frac{1}{\mathcal{R}_0} \log(S/S_0)$$
- Final size formula: $Z = 1 - e^{-\mathcal{R}_0 Z}$

The SIR model: Stability of equilibria

Phase Portrait ($\mathcal{R}_0 = 4$):



Model Equations:

$$\frac{dS}{dt} = -\beta SI, \quad \frac{dI}{dt} = \beta SI - \gamma I$$

Which equilibria are:

- Unstable?
 - $S_0 > 1/\mathcal{R}_0$
- Stable?
 - $S_0 \leq 1/\mathcal{R}_0$
- Asymptotically stable?
 - None!
- How do we prove these facts? (**Assignment 1**)

The SIR model: Effects of Control Measures

- How could the final size of the 1918 pandemic have been reduced?
- Any intervention that reduces $\mathcal{R}_0$ reduces the final size.
- What could have been done to reduce $\mathcal{R}_0$?
- Masks? Quarantine? Isolation?
- Ideally a vaccine, but no such luck in 1918.
- In 2009 (most recent flu pandemic), it took months to mass-produce a new vaccine. Even SARS-CoV-2 vaccine via mRNA took many months to develop and mass-produce.
- But suppose there had been a vaccine immediately...
- What proportion (p) of the population do we need to vaccinate to eradicate an infectious disease?

The SIR model: Effects of Control Measures

Suppose a proportion (p) of the population is vaccinated before an epidemic starts. Then:

- At the start of the epidemic, the proportion of the population that is susceptible is $S_0 = 1 - p$.
- $\therefore$ Initially (at time $t = 0$) the rate of change of prevalence is

$$\begin{aligned}\left.\frac{dI}{dt}\right|_{t=0} &= \left((\mathcal{R}_0 S - 1)I\right)\Big|_{t=0} = (\mathcal{R}_0 S_0 - 1)I_0 \\ &= (\mathcal{R}_0(1 - p) - 1)I_0 < 0 \iff \mathcal{R}_0(1 - p) < 1\end{aligned}$$

- $\therefore$ An epidemic will be prevented if

$$p > p_{\text{crit}} = 1 - \frac{1}{\mathcal{R}_0} \quad \text{\textit{herd immunity threshold}}$$

- $\therefore$ Public Health Agency will ask you to estimate $\mathcal{R}_0$.

The SIR model: Results so far

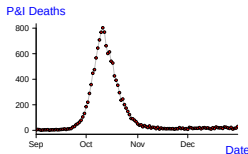
Biological inferences:

- $\mathcal{R}_0$ is extremely important to estimate in practice!
- Epidemic occurs if and only if $\mathcal{R}_0 > 1$.
- Single epidemic, then disease disappears.
 - Proof? **(Assignment 1...)**
Hint: **SIR phase portrait** indicates that every non-equilibrium solution is a heteroclinic orbit. (So every solution ends at an equilibrium.)
- Can prevent epidemic by vaccinating (or otherwise removing) a proportion $1 - \frac{1}{\mathcal{R}_0}$ from the transmission process.

Note: It doesn't matter whether we remove individuals from the susceptible pool by vaccination, isolation, or other means. What matters is the proportion of the population that is removed from the transmission process.

The SIR model: Does it explain our data?

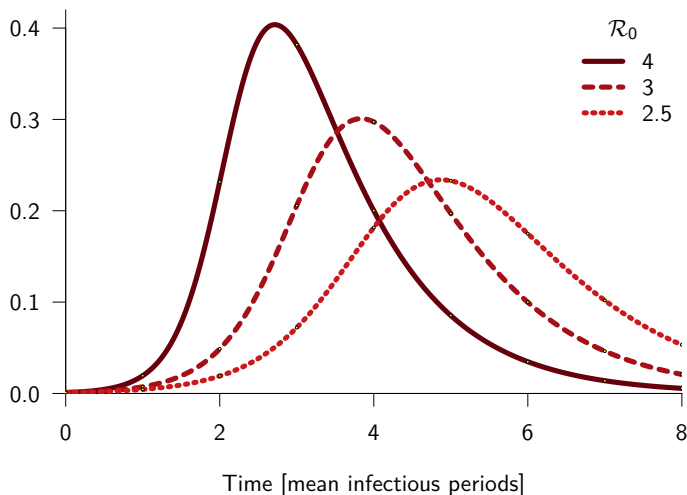
What about 1918 flu in Philadelphia?



- Does the SIR model explain these data?
- Can we fit the SIR model to the P&I time series?
- If so, are our estimated parameter values (for $\mathcal{R}_0$ and $1/\gamma$) biologically reasonable?
- Answers: **Assignment 2...**

The SIR model: How solutions depend on $\mathcal{R}_0$

Proportion infected $I(t)$



CPU time: 0.026S, Vector field evaluations: 1944, Ratio: 74769.2

The SIR model: prevalence vs. incidence

- In the **SIR model** as we have defined it, prevalence is $I(t)$ and incidence is

$$\iota(t) = \beta S(t)I(t),$$

so we can compute incidence $\iota(t)$ from solutions to the SIR model, and then compare predicted incidence with observed reports of cases or deaths.

- Could we have derived an equally sensible model starting from the closer-to-observable quantity (incidence ι) ?
- The answer is YES,

$$dS/dt = -\iota(t), \tag{1a}$$

$$\iota(t) = \mathcal{R}_0 S(t) \int_0^\infty \iota(t-s) g(s) ds, \tag{1b}$$

where $g(s)$ is the **generation interval distribution**.

- How do solutions of this integro-differential equation differ from those of the SIR model as we have defined it?

If you are curious, see [Champredon, Dushoff & Earn 2018](#).

L^AT_EX



T_EX and L^AT_EX

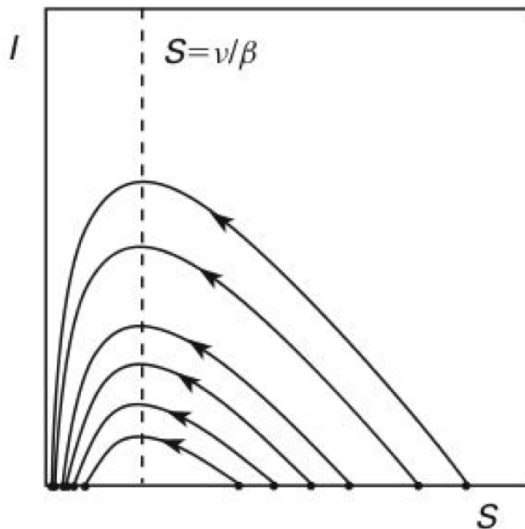
- T_EX is a computer-based typesetting system designed and implemented by Donald Knuth in the 1970s.
- L^AT_EX is a particular T_EX format written by Leslie Lamport. It has become the *de facto* standard for mathematical typesetting (e.g., books, journal articles, ...).
- T_EX has played an important role in the evolution of principles of software development.
 - Literate programming
 - Reproducible research
- Immediate goal: learn enough L^AT_EX to do Assignment 1.
- Goal for the term: become sufficiently competent with L^AT_EX and R so that the final project can be submitted as a fully reproducible document that “knits” L^AT_EX and R together.
- Work through LaTeX examples ...



Getting started

- Start RStudio
- Work by constructing an  script (see [Rexamples.R](#), accessible from [Lecture Schedule](#)).
- Try some arithmetic and basic plotting.
- Then try to make publication-quality graphs.
- Other resources:
 - Björnstadt “Epidemics: Models and Data Using R”, 2nd Ed (2023), SpringerLink
 - [Jonathan Dushoff's !\[\]\(924e710e3f6640277b0f86f71f23c067_img.jpg\) intro](#)
 - [Ben Bolker's !\[\]\(19762ac0eda9fbfb9176800dc8bc2fcd_img.jpg\) intro](#)
 - [R Project home page](#)
 - The official “Introduction to ” by Venables and Smith ([html](#), [pdf](#))
 - [Data Camp](#)
 - [Hadley Wickham's R for Data Science](#)

Figure 11.2 from HSD* (original from book)



*Hirsch, Smale and Devaney (2013), "Differential equations, dynamical systems, and an introduction to chaos".

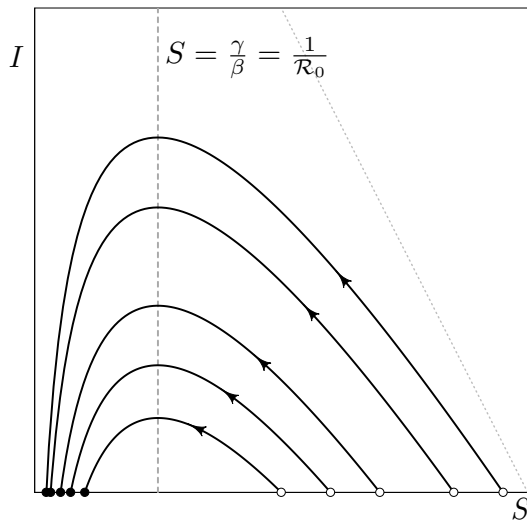
Figure 11.2 from HSD (made from scratch in )

Figure 11.1 from HSD (original from book)

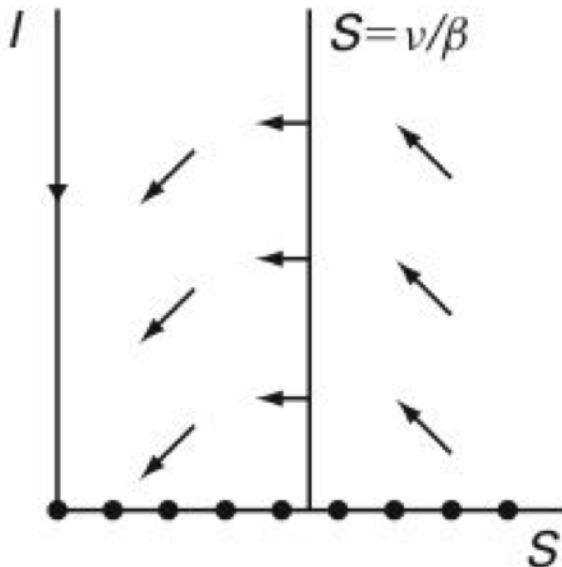


Figure 11.1 from HSD (made from scratch in )